Prior work on our theorem: exact tests under exchangeable errors

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Each section below takes one paper, states what it proves in its own mathematics, and then shows mathematically why it does not prove our theorem: whether it needs $p < n/2$ (Part 1) and whether it settles Part 2.

Bottom line.

- **Part 1 is confirmed.** Every method in the literature that is valid under exchangeable errors alone, for an arbitrary design, needs $p < n/2$ or something stricter. The methods that go further assume more about ε or need repeated design rows.
- **Part 2: $p < n/2$ is not necessary.** We give an elementary counterexample for a special design ($p = n - 3$). We also give a test that is valid for every p and, for almost every design, has power $\to 1$ at p well above $n/2$. That is proved rigorously at $n = 40$ ($p \le 25$) and $n = 20$ ($p \le 12$), with the general case stated as a conjecture.
- **What is necessary: $p \le n - 2$, for power uniform in β.** The Part 2 work is our own analysis and should be checked before it is presented.

The theorem on our board

This is the board, transcribed without additions.

Linear model.

$$Y_i = X_i^\top \beta + Z_i\, b + \varepsilon_i, \qquad i = 1, \dots, n, \qquad \beta,\, X_i \in \mathbb{R}^p, \qquad b,\, Z_i \in \mathbb{R}, \qquad \varepsilon_i \in \mathbb{R} \text{ exchangeable}$$

$$H_0 :\ b = 0 \qquad \text{vs} \qquad H_1 :\ b \neq 0, \qquad \beta \text{ a nuisance parameter}$$

The two regimes on the board: $p < n/2$, and $p = n/2$ (more generally $p \ge n/2$).

1. **Verify** whether all existing methods indeed need $p < n/2$ when ε is only exchangeable.
2. **Either** show that $p < n/2$ is necessary, **or** propose a method that works when $p \ge n/2$.

How the words on the board are read below. These are the definitions every paper in this review uses. They are stated so each paper can be judged against the same yardstick; they add nothing to the board.

- *Exchangeable* means $(\varepsilon_1, \dots, \varepsilon_n)$ has the same distribution as $(\varepsilon_{\pi(1)}, \dots, \varepsilon_{\pi(n)})$ for every permutation π. Nothing else is assumed: no independence, no symmetry, no moments.
- X_i, Z_i are treated as given numbers (a fixed design), as in every paper below.
- *A method that works* is a test φ with two properties:
  - **(V) validity:** for every n, every $\beta \in \mathbb{R}^p$ and every exchangeable law of ε,
  - **(P) power:** $P(\varphi \text{ rejects}) \to 1$ as $|b| \to \infty$. Some papers prove power in an asymptotic regime instead, which is noted where relevant.

$$\text{(V)} \qquad \mathbb{P}_{\beta,\, b=0}\big(\varphi \text{ rejects}\big) \ \le\ \alpha$$

Validity alone is not enough, because the test that ignores the data and rejects with probability α satisfies (V) for every p. Whether (P) must hold for every design X or for at least one design changes the answer to Part 2; the Part 2 section treats both readings.

Part 1 at a glance: does each method need $p < n/2$?

"Needs $p < n/2$" means the method is trivial or has no power once $p \ge n/2$. Here p counts the nuisance columns X; "p + 1" means the count includes Z. The details are in each paper's section.

#	Method	Assumption on ε	Design	Needs $p < n/2$?	Where it enters
1	RPT	exchangeable	any fixed	**Yes** (Theorem 2: $p < n/2$)	Each weight vector must kill X and $P_k X$: $n - 2p$ columns
2	PALMRT	exchangeable	any fixed	**Yes** (degenerate once $2p + 1 \geq n$); size $< 2\alpha$	Regression on [X, P_πX, Z] uses $2p + 1$ columns
3	Group PALMRT	exchangeable	any fixed	**Yes** (Theorem 1 assumes $p \leq n/2$)	Same as PALMRT
4	Fixed-row PALMRT	exchangeable	any fixed	**Yes**	Same as PALMRT
5	RobustPALMRT	exchangeable	any fixed	**Yes** (condition C1 forces a trivial test when $2p \geq n$)	C1
6	CPT	exchangeable	any fixed	**Stricter**: $p + 1 \leq \lfloor n/(m+1) \rfloor$ with $m + 1 = 1/\alpha$	Shift-invariant vectors
7	Pouliot exact t	exchangeable	any fixed	**Stricter**: about $n/12$ for $B = 4$ blocks	Span of the permuted copies gX
8	D'Haultfœuille–Tuvaandorj	exchangeable within strata	**repeated nuisance rows**	No, but trivial ($\varphi \equiv \alpha$) for a continuous design	Strata
9	Guo–Toulis–Wang	two-way exchangeable	any fixed	**Yes** ($p < N/2$)	$V_k \perp X$ and X_{π,σ}
10	Kolumbán et al. / RPS	exchangeable / i.i.d.	any fixed	Joint test only; its boundedness guarantee needs $p + 1 \leq n/2$	$Q_i \succ 0$
11	Fixed-X knockoffs	**Gaussian**	any fixed	**Yes** ($n \geq 2p$); beyond that only via Gaussianity	Constructing X
12	Huh–Jhun / Kherad-Pajouh–Renaud	**spherical**	any fixed	No, but invalid under exchangeability alone	$Q^T\varepsilon$ must be exchangeable
13	Freedman–Lane, Hemerik et al.	—	any	No guarantee for any p	—
14	DiCiccio–Romano	i.i.d., random design	random	Asymptotic, p fixed	—
15	Coudin–Dufour	**median-zero signs**	any fixed	Different assumption	—
16	Guo–Toulis	**sign-symmetric, independent**	any fixed	Partial null asymptotic only	—
17	Halkiewicz	**blockwise symmetric**	**dummy designs**	No, special designs only	Antithetic contrasts
18	Mosaic	local (panel) exchangeability	panel	Different problem	—
19–20	Dutz–Zhang; survey	—	—	Not applicable	—

Verdict on Part 1. The board's claim is **confirmed**, with one precise qualification. Every method that is (i) finite-sample valid under exchangeable errors alone and (ii) usable on an arbitrary fixed design needs $p < n/2$ or something stricter. That covers Papers 1–7 and 9, and the power guarantee of Paper 10. The methods that go past $n/2$ either assume more about ε (11, 12, 15–17) or need a special design with repeated rows (8, 17, and Paper 4's Remark 1).

Why the same number keeps appearing. The constructions that reach $n/2$ (Papers 1–5, 9 and 10) compare the data with **one** permuted copy at a time, and asks for directions that are orthogonal to both X and P_πX. That leaves $n - 2p$ free directions, which is positive only when $p < n/2$. (CPT and Pouliot use several copies, but they must cancel the nuisance in all copies at once, which uses up more dimensions and is stricter still.) The Part 2 section shows that comparing with **several** copies at once removes this barrier.

Paper 1 — Residual permutation test (RPT)

The paper

Kaiyue Wen, Tengyao Wang & Yuhao Wang, *Residual permutation test for regression coefficient testing*, Annals of Statistics 53(2), 2025. Its model (1) is our board's model, letter for letter: X ∈ ℝ^(n×p) nuisance, Z ∈ ℝⁿ tested, H_0: b = 0, fixed design.

What it proves

Assumptions. Assumption 1: ε is exchangeable. Assumption 2: the permutation matrices {$P_0 = I$, P_1, …, P_K} are closed under multiplication.

Construction.

1. Let $V_0 \in \mathbb{R}^{n\times(n-p)}$ have orthonormal columns with $V_0^{T}X = 0$, so V_0 is a basis of the residual space after regressing on X.
2. Let $V_k = P_k V_0$, the same basis with its rows permuted.
3. Let $\nabla_k \in \mathbb{R}^{n\times(n-2p)}$ have orthonormal columns lying both in the column space of V_0 and in the column space of V_k.

With T(x, y) = |⟨x, y⟩|, the p-value is

$$\phi = \frac{1}{K+1}\Big(1 + \sum_{k=1}^{K} 1\Big\{ \min_{1\le j\le K} T\big(\tilde V_j^{\top} Z,\ \tilde V_j^{\top} Y\big) \ \le\ T\big(\tilde V_k^{\top} Z,\ \tilde V_k^{\top} P_k Y\big)\Big\}\Big)$$

Theorem 2. If p < n/2, Assumptions 1–2 hold and b = 0, then P(φ ≤ α) ≤ α for every α ∈ [0, 1]. This holds for every n and every fixed design.

Proof.

1. Each column of ∇_k is a combination of the columns of V_0, so $\nabla_k^{T}X = 0$. Under H_0, $\nabla_k^{T}Y = \nabla_k^{T}(X\beta + \varepsilon) = \nabla_k^{T}\varepsilon$.
2. Each column of ∇_k is also a combination of the columns of P_kV_0, and $(P_kV_0)^{T}P_kX = V_0^{T}X = 0$. So $\nabla_k^{T}P_kY = \nabla_k^{T}P_k\varepsilon$. The unknown β has disappeared from both sides of every comparison.
3. Put g(a) = $\min_j$ $T(\nabla_j^{T}Z, \nabla_j^{T}a)$, a fixed function. Each right-hand side is at least $g(P_k\varepsilon)$, so
$$\phi \ \ge\ \phi' := \frac{1}{K+1}\Big(1 + \sum_{k=1}^{K} 1\{g(\varepsilon) \le g(P_k\varepsilon)\}\Big)$$
4. **Lemma 3.** Because $\{P_k\}$ is a group and $P_k\varepsilon$ has the same law as ε, the vector ($g(P_0\varepsilon)$, …, g(P_Kε)) is exchangeable. So the rank of g(ε) is (super-)uniform, with ties broken at random:
$$\mathbb{P}(\phi \le \alpha) \ \le\ \mathbb{P}(\phi' \le \alpha) \ \le\ \frac{\lfloor \alpha(K+1)\rfloor}{K+1} \ \le\ \alpha$$

Power (Theorem 3). Let Z = Xβ + e with ε and e i.i.d. and $E|\varepsilon_1|^{1+t} < \infty$. If n > (3 + m)p for a constant m > 0 and $|b| \gg n^{-t/(1+t)}$, then P(φ > 1/(K+1)) → 0. Theorems 9–10 show this rate is minimax-optimal up to n^{δ}.

Lemma 2 (the "naive" test). Permute the vector $V_0^{T}Y \in \mathbb{R}^{n-p}$ directly. This is valid for every p < n, but only if ε or (X, Z) is spherically symmetric. Section 9 offers this for n/2 ≤ p < n.

Why it does not prove our theorem

Part 1: yes, RPT needs p < n/2, and the paper says where. The construction needs ∇_k to have at least one column. A vector w lies in both column spaces exactly when

$$X^{\top}w = 0 \quad \text{and} \quad X^{\top}P_k^{\top}w = 0$$

That is 2p linear equations in n unknowns. In general they have a nonzero solution only when $n - 2p \geq 1$, and the paper's V_k has exactly $n - 2p$ columns. If $p \geq n/2$, then for a design in general position the only solution is $w = 0$. V_k is then empty, every T is 0, $\varphi = 1$, and the test never rejects. Our numerical check at $n = 40$: 2 usable directions at $p = 19$, none at $p = 20$.

Part 2: it proves neither branch.

- It gives a method only for $p < n/2$. Its only offering for $p \geq n/2$ is the naive test, which requires spherical symmetry. That is not on our board, and RPT's own Table 1 shows the naive test over-rejecting for t_1 noise.
- It never claims $p < n/2$ is necessary. The condition comes from the construction, not from a lower bound.

Why the naive test fails under mere exchangeability: it needs $V_0^T \varepsilon$ to be exchangeable in $\mathbb{R}^{n-p}$. Spherical symmetry passes through V_0^T, but exchangeability does not, because V_0^T is not a permutation.

Paper 2 — Permutation-augmented regression test (PALMRT)

The paper

Leying Guan, *A conformal test of linear models via permutation-augmented regressions*, Annals of Statistics 52(5), 2024. Translation: Guan's x is our Z (tested), and Guan's $Z \in \mathbb{R}^{n \times p}$ with coefficient θ is our X with coefficient β.

What it proves

Model. $y = x\beta + Z\theta + \varepsilon$, ε exchangeable (Assumption 1.1), H_0: $\beta = 0$. For a permutation π, x_π and Z_π are the design with rows permuted. H^A denotes the projection (hat) matrix onto the columns of A.

Statistic. For B independent uniform permutations π_b, run two least-squares regressions of y, each on $2p + 1$ columns, and record the residual sums of squares:

$$T_{0b} = \left\|(I - H^{x_{\pi_b}\, Z\, Z_{\pi_b}})\, y\right\|_2^2 \quad (\text{permuted } x), \qquad T_{b0} = \left\|(I - H^{x\, Z\, Z_{\pi_b}})\, y\right\|_2^2 \quad (\text{true } x)$$

$$p_{\text{val}} = \frac{1 + \sum_{b=1}^{B} \omega_b}{B + 1}, \qquad \omega_b = 1\{T_{b0} > T_{0b}\} + \tfrac{1}{2} 1\{T_{b0} = T_{0b}\}$$

If $\beta \neq 0$, the regression containing the true x fits better, so $T_{b0} < T_{0b}$ and p_{val} is small.

Theorem 3.3. Under H_0, $P(p_{val} \leq \alpha) < 2\alpha$ for every $\alpha > 0$, for any statistic T satisfying Condition 3.1 below. The theorem has no condition on p.

$$T(\pi_1, \pi_2; \varepsilon_\sigma) = T(\pi_1 \circ \sigma^{-1},\ \pi_2 \circ \sigma^{-1};\ \varepsilon) \qquad \forall\, \pi_1, \pi_2, \sigma$$

Proof.

1. Under H_0, $y = Z\theta + \varepsilon$, and both regressions contain Z, so θ drops out. With $\pi_0 = \text{id}$, $T_{0b} = T(\pi_0, \pi_b; \varepsilon)$ and $T_{b0} = T(\pi_b, \pi_0; \varepsilon)$, where $T(\pi_1, \pi_2; \varepsilon) = \|(I - H^{(x_{\pi_2}\, Z_{\pi_2}\, Z_{\pi_1})})\varepsilon\|^2$ satisfies Condition 3.1.
2. Draw an independent uniform σ. Then σ^{-1}, $\pi_1 \circ \sigma^{-1}$, …, $\pi_B \circ \sigma^{-1}$ are again i.i.d. uniform (Prop. 5.1). With $\varepsilon_\sigma =_d \varepsilon$, the whole array $T(\pi_l, \pi_k)$ has the law of one built from i.i.d. uniform permutations, so index 0 is exchangeable with the others.
3. Tournament count (Prop. 5.2). Put $W_{lk} = \mathbb{1}\{T_{kl} > T_{lk}\} + \tfrac{1}{2}\mathbb{1}\{T_{kl} = T_{lk}\}$ and $W_{ll} = 1$, so $W_{lk} + W_{kl} = 1$. If s rows each have row sum $\leq \alpha(B+1)$, their $s \times s$ block sums to $s(s+1)/2$, which gives the bound below.
4. By step 2, P(row 0 is one of those rows) $= E(s)/(B+1) < 2\alpha$.

$$\frac{s(s+1)}{2} \leq s\,\alpha(B+1) \implies s < 2\alpha(B+1)$$

Why it does not prove our theorem

Part 1: it is valid for every p, but it can only reject when $2p + 1 < n$, i.e. $p < (n-1)/2$. Each regression uses the $2p + 1$ columns (x or x_π, Z, Z_π). For a design in general position these columns span all of $\mathbb{R}^n$ once $2p + 1 \geq n$, so both fits are perfect:

$$2p+1 \geq n \implies T_{0b} = T_{b0} = 0 \ \forall b \implies \omega_b = \tfrac{1}{2} \implies p_{\text{val}} = \frac{1+B/2}{B+1} \geq \frac{1}{2}$$

So it never rejects at any α < 1/2. For odd n this is slightly stricter than p < n/2: at p = (n − 1)/2 it already fails. Our check at n = 40 gives a residual norm of 0.37 at p = 19 and 6×10^{-14}, i.e. zero, at p = 20. Guan's Table 1 lists the restriction "n > 2p".

Part 2: it proves neither branch.

- Its guarantee is 2α, not α. Paper 3 proves the 2 cannot be removed for this construction.
- It gives no method for p ≥ n/2, since there the test is constant.
- It gives no lower bound showing p < n/2 is necessary. Its only stated open question is whether filtering permutations improves power.

Paper 3 — Group PALMRT

The paper

Zonghan Li, Hongyi Zhou & Zhiheng Zhang, *Group permutation testing in linear model: sharp validity, power improvement, and extension beyond exchangeability*, arXiv:2601.17734 (January 2026, revised April 2026). Translation: their Y = Zβ + bX + ε has Z ∈ ℝ^(n×p) as nuisance and X ∈ ℝⁿ tested.

What it proves

Construction. PALMRT's comparison, with the permutations {P₀ = I, …, P_K} required to form a group (Assumption 2):

$$\phi = \frac{1}{K+1}\Big(1 + \sum_{k=1}^{K} 1\big\{X^\top (I - H^{ZZ_{\pi_k}})Y \ \leq \ X_{\pi_k}^\top (I - H^{ZZ_{\pi_k}})Y\big\}\Big)$$

Theorem 1. Let p ≤ n/2, ε exchangeable, {P_k} a group and b = 0. Let S = {m : R_m ≤ α} be the indices with small rank statistic. Then

$$\mathbb{P}(\phi \leq \alpha) \leq \frac{\mathbb{E}|S|}{K+1}, \qquad \frac{\lfloor \alpha(K+1) \rfloor}{K+1} \leq \frac{|S|}{K+1} \leq \frac{\lfloor 2\alpha(K+1) \rfloor}{K+1}$$

Proposition 2. For every n ≥ 2p there are a design, a group and an α with P(φ′ ≤ α) = 2α for every noise vector. So the factor 2 cannot be improved without extra assumptions.

Theorems 10–11. Without exchangeability, validity degrades by the average total-variation distance:

$$\mathbb{P}\Big(R_0 \leq Q_{1-\alpha}\big(\textstyle\sum_k \frac{1}{K+1}\delta_{R_k}\big)\Big) \ \geq \ 1 - \alpha - \frac{1}{K+1}\sum_{k=1}^{K} d_{TV}\big(\varepsilon, \ \varepsilon_{\pi_k}\big)$$

Proof of Theorem 1.

1. Under H₀, I − H^(Z Z_π_k) removes both Zβ and Z_π_kβ, so every comparison depends on ε alone.
2. Prop. 1 (equivariance): F(π₁, π₂; ε_σ) = F(σ⁻¹∘π₁, σ⁻¹∘π₂; ε).
3. Prop. 4: multiplying by any P_k permutes the group. With exchangeability, R₀, …, R_K are identically distributed.
4. Prop. 5: P(R₀ ∈ S) = (1/(K+1)) Σ_k P(R_k ∈ S) = E|S|/(K+1). PALMRT's tournament count gives |S| ≤ ⌊2α(K+1)⌋.

Why it does not prove our theorem

Part 1: yes, Theorem 1 assumes p ≤ n/2, and at p = n/2 the test cannot reject. Generically Z and Z_π together have rank min(n, 2p). At p = n/2 that is n, so H^(Z Z_π) = I and both sides of every comparison are 0. Then φ = 1. The boundary is included only because validity is trivial there.

Part 2: it proves neither branch. "Sharp" refers to the constant 2 in the size bound at a fixed p ≤ n/2, not to the dimension. Theorems 10–11 relax the assumption on ε rather than on p. The paper's open questions are the factor 2 (settled by Prop. 2), Type II error, and going beyond exchangeability. Its conclusion mentions "high-dimensional"

problems only as a vague future direction.

A correction to the paper. Theorem 9 states that group-CPT's linear system has a nonzero solution iff n/p > K. The "only if" half is false: $\eta^* = \mathbf{1}$ always solves it, because $\mathbf{1}$ is unchanged by every permutation. Paper 6 explains why that solution is useless.

Paper 4 — Fixed-row PALMRT

The paper

Tianyi Wang, Guanghui Wang, Zhaojun Wang & Changliang Zou, *Power enhancement of permutation-augmented partial-correlation tests via fixed-row permutations*, Statistics & Probability Letters 2025 (doi 10.1016/j.spl.2025.110591). Notation as in PALMRT: X tested, Z ∈ ℝ^(n×p) nuisance.

What it proves

Construction. A greedy algorithm chooses a row set $\mathscr{B} = \mathscr{B}(X, Z)$ to hold fixed. PALMRT is then run with permutations drawn uniformly from the subgroup that leaves those rows in place. The aim is to avoid the event X ∈ span(Z, P_πZ), which kills power.

Theorem 1. Under PALMRT's Condition 1, for any fixed design, any exchangeable ε and any $\mathscr{B}(X, Z)$:

$$\mathbb{P}\big(p_{\text{val}} \le \alpha \mid H_0\big) < 2\alpha$$

Proof: the permutations fixing $\mathscr{B}$ form a subgroup of S_n, and Guan's argument (Paper 2) runs unchanged inside it.

Remark 1. If every allowed permutation leaves the nuisance unchanged, P_(π|$\mathscr{B}$)Z = Z, then the size is at most α. In that case

$$T_{i0} = \|(I - H_{X,Z})Y\|_2^2, \qquad T_{0i} = \|(I - H_{X,Z})P_{\pi_i}^{-1}Y\|_2^2,$$

and under H_0 the T's are exchangeable.

Appendix A.3. In a paired design, fixing any single row still leaves P(X ∈ span(Z, P_πZ)) ≥ (p − 1)/(n − 1), which is ≥ α once p ≥ α(n − 1) + 1.

Why it does not prove our theorem

Part 1: it inherits PALMRT's requirement. Fixing rows does not reduce the number of regression columns, which is still 2p + 1. The perfect-fit argument of Paper 2 applies unchanged: the test cannot reject once 2p + 1 ≥ n. The paper states no dimension condition of its own and attributes "n > 2p" to PALMRT.

Part 2: it proves neither branch, with one instructive exception. Remark 1's exact case needs P_πZ = Z for nontrivial permutations, which happens only if the rows being permuted are identical in Z. For such designs the 2p + 1 count drops to p + 1, so the method keeps power past n/2. But it is the same special-design phenomenon as Paper 8, not a method for designs whose rows are all different.

Paper 5 — RobustPALMRT

The paper

Torey Hilbert, Steven MacEachern & Yuan Zhang, *Robust distribution-free tests for the linear model*, Statistics in Medicine 2026. Notation: Y = Xβ + Zθ + ε, with X tested and Z ∈ ℝ^(n×p) nuisance.

What it proves

Construction. Replace least squares by any fitting-and-scoring procedure $\mathscr{M}$ (Huber regression with MAD scale, quantile regression, dispersion fits). Then compare the score using the true X with the score using X_πb:

—

$$A_b = 1\{\mathcal{M}(Y, X, [Z, Z_{\pi_b}]) \;\geq\; \mathcal{M}(Y, X_{\pi_b}, [Z, Z_{\pi_b}])\}, \qquad \mathfrak{p} = \frac{1 + \sum_b A_b}{B + 1}$$

Conditions. $\mathscr{M}$ must satisfy, for all $\gamma \in \mathbb{R}^{2p}$ and all σ:

$$\text{(C1)}\;\; \mathcal{M}(Y + [Z, Z_\pi]\gamma,\, X,\, [Z, Z_\pi]) = \mathcal{M}(Y, X, [Z, Z_\pi]), \qquad \text{(C2)}\;\; \mathcal{M}(Y_\sigma, X_\sigma, [Z, Z_\pi]_\sigma) = \mathcal{M}(Y, X, [Z, Z_\pi])$$

Theorem 1. If $\mathscr{M}$ satisfies (C1)–(C2), ε is exchangeable and Y = Zθ + f(X, ε), then P_H$_0$(p ≤ α) ≤ 2α for every α and θ. Theorem 2 extends this to non-uniform permutation distributions and subgroups. Lemma 1 shows that Huber regression satisfies (C1)–(C2).

Proof idea: (C1) removes θ, since Zθ is a combination of the columns of [Z, Z_π]. (C2) gives PALMRT's equivariance, and the tournament count gives 2α.

Why it does not prove our theorem

Part 1: power needs 2p < n. (C1) says the score is unchanged when Y moves by any combination of the 2p columns of [Z, Z_π]. For a design in general position those columns span $\mathbb{R}^n$ once 2p ≥ n. Every Y can then be moved to 0, so

$$\mathcal{M}(Y, \cdot, \cdot) = \mathcal{M}(0, \cdot, \cdot) \quad \text{for every } Y$$

The score ignores the data and the test cannot have power.

Part 2: it proves neither branch. The authors note that Z is limited to modest dimension. They say a high-dimensional version is possible with a known bound on $\|\theta\|$, but has low power and is left out. That bound is an extra assumption: our β is unrestricted. The guarantee is also 2α, not α.

Paper 6 — Cyclic permutation test (CPT)

The paper

Lihua Lei & Peter J. Bickel, *An assumption-free exact test for fixed-design linear models with exchangeable errors*, Biometrika 108(2), 2021. It was the first test exact under exchangeable errors alone for an arbitrary fixed design. Translation: their y = β$_0$**1** + Xβ + ε has X ∈ ℝ^(n×p) containing the tested column(s) and the nuisance. For us r = 1 tested column, so the nuisance has p − 1 columns plus a free intercept.

What it proves

Idea. Build m + 1 linear statistics S$_j$ = y^{T}η$_j$ whose joint law under H$_0$ is unchanged by a cyclic shift. The rank of S$_0$ among them is then uniform and gives an exact p-value.

Construction. Choose m (m = 1/α − 1 for an exact level-α test) and block size t = ⌊n/(m+1)⌋. Π$_j$ shifts the first (m+1)t coordinates by j blocks of length t and leaves the rest fixed. The weight vectors must satisfy

$$\eta_j = \Pi_j \eta^* \quad \text{(C2)}, \qquad X_{[-r]}^\top \eta_j = \gamma \;\text{ for all } j \quad \text{(C1)}, \qquad S_j = y^\top \eta_j$$

(C1)–(C2) form a homogeneous linear system (11) in (γ, η*): (m+1)(p − r) equations in n + p − r unknowns.

Proposition 1. If (S$_0$, …, S$_m$) has the same law as each cyclic shift, then the descending rank R$_0$ of S$_0$ satisfies

$$\mathbb{P}(R_0 \geq k) = \frac{1}{m+1}\,\mathbb{E}\big|\{j : R_j \geq k\}\big| \;\geq\; \frac{m - k + 2}{m + 1}$$

Proposition 2. p = R$_0$/(m+1) is valid. It is exactly level α if 1/α is an integer, (m+1) is divisible by 1/α, and there are no ties.

Theorem 1. Under exchangeability: (a) system (11) has a nonzero solution if n/(p − r) > m; (b) for any solution, (S$_0$, …, S$_m$) is invariant under cyclic shifts when H$_0$ holds. Proof of (b):

$$S_j = \underbrace{(\mathbf{1}^\top \eta_j)\beta_0 + \gamma^\top \beta_{[-r]}}_{\text{the same for every } j} + \varepsilon^\top \Pi_j \eta^*$$

$\mathbf{1}^{\mathsf{T}}\eta_j$ is the same for every j because η_j is a rearrangement of η^*. Since $\Pi_{j+1} = \Pi_j\Pi_1$ and $\Pi_1{}^{\mathsf{T}}\varepsilon$ has the same law as ε, the vector $(\varepsilon^{\mathsf{T}}\Pi_j\eta^*)_j$ has the same law as its cyclic shift.

Theorem 2 (power). To separate S_0 from the others under H_1, choose η to maximise the signal $X_1{}^{\mathsf{T}}(\eta_0 - \eta_m)$ subject to (C1). With $B(X) = ((I - \Pi_m)^{\mathsf{T}}X, (\Pi_1 - \Pi_m)^{\mathsf{T}}X, \ldots, (\Pi_{m-1} - \Pi_m)^{\mathsf{T}}X)$, the optimum is the least-squares residual of B(X)'s first column on its other columns: $\tilde{\eta} = (I - H_[-1])B(X)_1$.

Why it does not prove our theorem

Part 1: yes, and its requirement is much stricter than $p < n/2$. With $m = 1/\alpha - 1$, the stated condition $n/(p - r) > m$ reads

$$p - r \;<\; \frac{n}{m} = \frac{n\,\alpha}{1-\alpha} \qquad (\alpha = 0.05:\;\; p - r < n/19)$$

The condition actually needed for power is stricter still (our derivation, confirmed independently).

- Theorem 1(a) is automatically true: any η^* that every shift leaves unchanged solves (11), e.g. $\eta^* = \mathbf{1}$. Such an η^* makes $S_0 = S_1 = \ldots = S_m$, so the test never rejects.
- Power needs $\tilde{\eta} \neq 0$, i.e. $B(X)_1$ must not be a combination of the other columns of B(X).
- Take a shift-invariant vector u: equal on all m+1 blocks, arbitrary on the tail. There are t + s free values, where $n = (m+1)t + s$. Every column of B(X) is orthogonal to every such u, because $(\Pi_j - \Pi_m)u = u - u = 0$.
- So the mp columns of B(X) satisfy t + s independent linear restrictions and live in a space of dimension mt. In general position $B(X)_1$ avoids the others' span only if $mp - 1 < mt$:

$$\tilde{\eta} \neq 0 \quad\Longleftrightarrow\quad p \le \left\lfloor \frac{n}{m+1} \right\rfloor \quad (p \text{ counts the tested column})$$

The independent check confirmed this numerically for many (n, m). At $n = 100$ and $\alpha = 0.05$, power needs at most 4 nuisance columns, while the paper's condition allows 5.

Even its most permissive setting stops at n/2. Take $m = 1$, so $\alpha = 1/2$ and Π_1 swaps the two halves. The formula gives $p \le n/2$ counting the tested column, i.e. fewer than n/2 nuisance columns. So CPT cannot have power with $p \ge n/2$ at any level.

Part 2: it proves neither branch. It gives no method for $p \ge n/2$ and no claim that its condition is necessary. Its only remark is that for $p > n$, inference would need extra assumptions such as sparsity.

Paper 7 — An exact t-test with block permutations

The paper

Guillaume A. Pouliot, *An Exact t-Test*. We read the 2024 working paper. A replication package dated March 2026 sits in the Econometric Society's Zenodo community, indicating a 2026 journal version. Translation: X_1 is our Z, and X_2 together with the intercept is our nuisance X.

What it proves

Model. $Y = \mathbf{1}\beta_0 + X_1\beta_1 + X_2\beta_2 + \varepsilon$, testing H_0: $\beta_1 = \beta_1{}^0$.

Construction. G is a group of permutations; the recommended choice is block permutations. Split the n rows into B blocks and permute whole blocks, so $|G| = B!$. Let Q be the projection that removes every permuted copy gX_2, i.e. projection onto the complement of $\mathrm{span}\{gX_2 : g \in G\}$. Residuals $\hat{\varepsilon}$ come from regressing on $\mathbf{1}$, $\{gX_1\}$ and $\{gX_2\}$. Then

$$t_g(\beta_1^0) = \frac{\bar{X}_1^{\top}\, g\,(Y - X_1\beta_1^0)}{\hat{\sigma}_g}, \qquad \bar{X}_1 = QX_1, \qquad \hat{\sigma}_g^2 = \frac{1}{n}\sum_{i=1}^{n} \bar{X}_{i1}^2\, \hat{\varepsilon}_{g(i)}^2$$

Reject when the observed t exceeds the k-th order statistic of $\{t_g : g \in G\}$, with $k = M - \lfloor M\alpha \rfloor$, randomising at ties.

Theorem 1. Suppose $(X_1, X_2, \varepsilon) =_d (X_1, X_2, g\varepsilon)$ for all $g \in G$ (true for exchangeable ε and a fixed design), and $Y - X_1\beta_1^0 \notin \text{span}\{g(X_1, X_2)\}$. Then $E\varphi = \alpha$ exactly.

Proof.

1. The set $\{gX_2\}$ is mapped to itself by every g, so Q commutes with g (Lemma A.1).
2. Under H_0, the numerator is $X_1^T Q g(X_2\beta_2 + \varepsilon) = X_1^T Q g\varepsilon$, since $QgX_2 = 0$. The residuals satisfy $\hat{\varepsilon}(g) = g\hat{\varepsilon}(\text{id})$. So t_g depends on the data only through $g\varepsilon$.
3. G is a group, so $\Sigma_g \varphi(gW) = M\alpha$ for every data set. Each $\varphi(gW)$ has the law of $\varphi(W)$, hence $E\varphi = \alpha$.

Theorem 2 / A.1. If $n \geq p \cdot B^2$, then

$$\max_{X \in \mathbb{R}^{n \times p}} \operatorname{rank}\left(g^{(1)}X, \ldots, g^{(M)}X\right) = p \cdot \big(B(B-2)+2\big)$$

Why it does not prove our theorem

Part 1: yes, and the requirement is far stricter than $p < n/2$. Power needs $QX_1 \neq 0$, so X_1 must not be a combination of the permuted copies gX_2. Those copies span up to $(p-1)(B(B-2)+2)$ dimensions by Theorem A.1, so n must exceed that. An exact level-0.05 test needs $B! \geq 20$, so $B \geq 4$:

$$B = 4: \quad \operatorname{rank}\{gX_2\} \approx (p-1)\cdot 10 \quad \Longrightarrow \quad \text{need roughly } n > 10(p-1)$$

The independent check put the generic threshold near $p - 1 < n/(B(B-1)) = n/12$. Either way it is far below $n/2$. Even the smallest group, $B = 2$, gives rank $2(p-1)$, so it needs $p - 1 < n/2$, and then only $\alpha = 1/2$ is available.

Part 2: it proves neither branch. Its contribution is exactness *plus* asymptotic robustness to heteroskedasticity, not dimension.

A discrepancy to check. The paper reports simulated power at $p = 35$, $n = 50$, $B = 5$. There Theorem A.1 gives a span far larger than 50 dimensions (35×17), and the independent check found $QX_1 = 0$ for generic designs. We could not reconcile this. Check the published version before citing it.

Paper 8 — Stratified (within-strata) permutation test

The paper

Xavier D'Haultfœuille & Purevdorj Tuvaandorj, *A robust permutation test for subvector inference in linear regressions*, Quantitative Economics 15(1), 2024. Translation: their tested regressors X (k columns) are our Z ($k = 1$). Their nuisance Z (p columns, including the intercept) is our X. We write everything below in **our** letters.

What it proves

Model. $Y = X\beta + Zb + \varepsilon$, testing H_0: $b = b_0$.

Strata. A stratum is a set of rows that share the **same nuisance row**: $X_i = X_j$. Let there be S strata of sizes $n_1, \ldots, n_S$. A *stratified permutation* π shuffles rows only inside strata, so

$$P_\pi X = X \quad \text{for every stratified } \pi, \qquad |\mathbb{S}_n| = \prod_{s=1}^{S} n_s!$$

Statistic. Let D be within-stratum demeaning and $\tilde{Z} = DZ$. Then

$$W(v) = \frac{(v^\top \tilde{Z})^2}{\sum_i \tilde{Z}_i^2 \, [Dv]_i^2}, \qquad v = Y - Zb_0, \qquad \text{reject if } W(v) \text{ exceeds the } (1-\alpha) \text{ quantile of } \{W(P_\pi v) : \pi \in \mathbb{S}_n\}$$

(ties are broken by randomisation).

Assumption 1. (a) ε is exchangeable *within each stratum*; (b) Z and ε are independent given X; (c) [X, Z] has full column rank.

Theorem 3.1 of the arXiv version ("Theorem 1" in the journal). Under H_0 and Assumption 1, $E[\varphi \mid X, Z] = \alpha$ for every α, with **no restriction on p**. The test is randomised at ties.

Proof in our notation. Under H_0, $v = X\beta + \varepsilon$. X is constant within strata, so $DX = 0$, and $P_\pi X = X$:

$$W(P_\pi v) = W(X\beta + P_\pi \varepsilon) = W(P_\pi \varepsilon) \quad (W \text{ depends on } v \text{ only through } Dv \text{ and } v^\top \tilde{Z}, \text{ and } \tilde{Z}^\top X = 0)$$

Within-stratum exchangeability gives $P_\pi \varepsilon =_d \varepsilon$, so $\{W(P_\pi v)\}$ is invariant under the group $\mathbb{S}_n$. The standard group argument gives exactness.

Why it does not prove our theorem

Part 1: this method does *not* need $p < n/2$ — but it only works for a special kind of design. Nothing in Theorem 3.1 involves p versus n. The price is the design.

- If no two rows of X are equal (any continuous design, the generic case on our board), every stratum has size 1. Then $\mathbb{S}_n = \{\text{identity}\}$ and the test is the trivial $\varphi \equiv \alpha$. The paper says this explicitly.
- If X has S distinct rows, every column of X is a combination of the S stratum indicators, so $\text{rank}(X) \le S$. Power needs $Z \neq 0$ (Z varies inside some stratum) and $\prod n_s! \ge 1/\alpha$.
- So it covers $p \ge n/2$ only when the nuisance design has *repeated rows*.

Part 2: it gives the literature's counterexample for special designs. It does not claim this. Matched pairs ($S = n/2$, X = pair indicators, $p = n/2$) is a valid test with power. A design with one stratum of size 4 and $n - 4$ singletons has $p = n - 3$ and still works at $\alpha = 0.05$, because $4! = 24 \ge 20$. The full proof is in the Part 2 section below. It does not give a method for a design without repeated rows, and it does not show necessity.

Caveat about the studentised statistic. As $|b| \to \infty$ the observed W tends to $\|Z\|^4 / \Sigma Z_i^4$, and this need not beat every permuted value. Example: $Z = (2, 1, -1, -2)$ in a stratum of size 4 gives $100/34 \approx 2.94$ observed, but 4 for the permutation sending it to $(1, 2, -2, -1)$. That is why our Part 2 example uses the unstudentised statistic $|Z^T v|$.

Paper 9 — Permutation inference under multi-way clustering

The paper

Wenxuan Guo, Panos Toulis & Yuhao Wang, *Permutation Inference under Multi-way Clustering and Missing Data*, arXiv, January 2026. It extends RPT (Paper 1) to dyadic and panel data.

What it proves

Model (7). One observation per cell of an $n \times n$ array, $N = n^2$ observations in total:

$$y_{ij} = x_{ij}^\top \gamma + d_{ij}^\top \beta + \varepsilon_{ij}, \qquad y = X\gamma + D\beta + \varepsilon, \qquad H_0 : \beta = 0$$

Here $\gamma \in \mathbb{R}^p$ is the nuisance. In our letters: X ↔ our X, D ↔ our Z.

Assumption 1 (separate exchangeability). For all row permutations π and column permutations σ, $(\varepsilon_{ij}) =_d (\varepsilon_{\{\pi(i)\sigma(j)\}})$ given X, D.

Procedure 1. Draw K pairs (π_k, σ_k) forming a group G. Choose an orthonormal $V_k \in \mathbb{R}^{N \times (N-2p)}$ with

$$V_k^\top X = 0, \qquad V_k^\top X_{\pi_k, \sigma_k} = 0, \qquad a_k = \|D^\top V_k V_k^\top y\|, \qquad b_k = \|D^\top V_k V_k^\top y_{\pi_k, \sigma_k}\|$$

$$\text{pval} = \frac{1}{K+1}\Big(1 + \sum_k \mathbf{1}\{\min_j a_j \le b_k\}\Big)$$

Theorem 1. If $p < N/2$ and Assumption 1 holds, then under H_0, $P(\text{pval} \le \alpha \mid X, D) \le \alpha$ for all α.

Why it does not prove our theorem

Part 1: yes, it requires p < N/2 — the same requirement as RPT. The condition is in the statement of Theorem 1. It enters because V_k must kill both X and its permuted copy, which is 2p linear conditions on an N-dimensional vector:

$$\dim\{v:\ v^\top X = 0,\ v^\top X_{\pi,\sigma} = 0\}\ \geq\ N - 2p > 0\ \iff\ p < N/2$$

The error model is also different from ours. Separate exchangeability on an array is not exchangeability of a vector. With one cluster dimension, the procedure reduces to RPT.

Part 2: it proves neither branch. It neither treats p ≥ N/2 nor shows the condition is necessary.

Paper 10 — Exact joint confidence regions from permuted residuals (perturbed datasets, RPS)

The papers

- Sándor Kolumbán, István Vajk & Johan Schoukens, *Perturbed datasets methods for hypothesis testing and structure of corresponding confidence sets*, arXiv, December 2014.
- Szabolcs Szentpéteri & Balázs Csanád Csáji, *Finite-sample identification of linear regression models with residual-permuted sums*, arXiv, 2024.

Both come from system identification and test the **whole** parameter vector at once. In our notation the whole vector is θ = (β, b) ∈ ℝ^(p+1), with Ψ = [X, Z]. Let H = Ψ(Ψ^T Ψ)^{-1} Ψ^T, M = I − H, and r(θ) = Y − Ψθ.

What they prove

Kolumbán–Vajk–Schoukens, exchangeable version. Take $P_0 = I$ and P_1, …, P_K uniform random permutations. Define

$$Z_i(\theta) = r(\theta)^\top P_i^\top H P_i\, r(\theta) = \|H P_i\, r(\theta)\|^2, \qquad i = 0, \dots, K$$

Accept θ unless $Z_0(\theta)$ is among the q largest values. At the true θ*, r(θ*) = ε, so (Z_0, …, Z_K) is exchangeable when ε is exchangeable. The rank of Z_0 is then uniform, which gives

$$\mathbb{P}\big(\theta^* \notin C\big) \leq \frac{q}{K+1} = \alpha \qquad \text{for every } n,\ p \text{ and every exchangeable law of } \varepsilon$$

Because $\|P_i r\| = \|r\|$ and $\|Mr(\theta)\| = \|MY\|$ for every θ, the acceptance rule can be rewritten as

$$\theta \in C\ \iff\ \#\{\, i \geq 1 :\ \|M P_i\, r(\theta)\| \leq \|MY\| \,\}\ \geq\ q$$

Their Theorem 4 (structure). The region is connected. It is **bounded** if the design is *sufficiently exciting* with respect to the non-identity permutations used. Their condition is Q > 0, with their X ∈ ℝ^(n_θ×n) equal to our Ψ^T; in our notation it reads

$$Q_i = \Psi^\top \Psi - \Psi^\top P_i^\top H P_i \Psi = (P_i\Psi)^\top M\,(P_i\Psi)\ \succ\ 0$$

Szentpéteri–Csáji (RPS). Define the reference sum $S_0(\theta) = R^{-1/2}\, n^{-1}\, \Sigma_t\, \psi_t\, \varepsilon_t(\theta)$ and the permuted sums $S_i(\theta)$, where ε_t is permuted. Their Theorem 1: if the noise is **i.i.d.** and independent of the instruments, coverage is exactly 1 − q/m for every n. Their Theorem 2: strong consistency as n → ∞ with the dimension fixed.

Why they do not prove our theorem

Part 1: they are joint tests, and their power guarantee needs p < n/2.

- They test all p + 1 coefficients jointly. Our H_0 leaves β free, which these papers never address. A test of b = 0 can be made from C by **projection**: reject when no θ = (β, 0) lies in C. That test is valid for every p (see the Part 2 section), but neither paper studies it.

- Their only guarantee that C is bounded, which is what power needs, is $Q_i \succ 0$. That requires M P_i Ψ to have full column rank p + 1, but M has rank n − p − 1:

$$Q_i \succ 0 \implies p+1 \le n-p-1 \iff p \le \tfrac{n}{2}-1 \ \ (< n/2)$$

The n/2 barrier therefore reappears inside their boundedness theorem. When p ≥ n/2, every Q_i is singular, and each single set $\{\theta : \|MP_ir(\theta)\| \le \|MY\|\}$ is an unbounded cylinder.

- RPS additionally assumes i.i.d. noise, and its consistency theory has the dimension fixed.

Part 2: they prove neither branch, but this is where our candidate method comes from. Acceptance needs **q** cylinders to hold at the same time. An intersection of q unbounded cylinders can still be bounded when p ≥ n/2. That observation is the basis of the method in the Part 2 section.

Paper 11 — Fixed-X knockoffs

The paper

Rina Foygel Barber & Emmanuel J. Candès, *Controlling the false discovery rate via knockoffs*, Annals of Statistics 43(5), 2015.

What it proves

Model. y = Xβ + z with **z ~ N(0, σ²I)**, a fixed design X ∈ ℝ^(n×p), and **n ≥ 2p**. Let Σ = X^TX.

Knockoffs. Construct $\tilde{X}$ with

$$\tilde{X}^\top \tilde{X} = \Sigma, \qquad X^\top \tilde{X} = \Sigma - \operatorname{diag}(s), \qquad \tilde{X} = X(I - \Sigma^{-1}\operatorname{diag}(s)) + \tilde{U}C$$

Here $\tilde{U}$ ∈ ℝ^(n×p) is orthonormal and orthogonal to the columns of X. That is only possible when n ≥ 2p.

Theorems 1–2. The knockoff+ procedure controls the false discovery rate at level q exactly in finite samples. The proof uses Gaussianity: swapping X_j and $\tilde{X}_j$ for null j leaves the joint law of the statistics unchanged.

Section 2.1.2 (p < n < 2p). Append 2p − n zero rows to X, and append 2p − n synthetic responses drawn from N(0, $\hat{\sigma}^2$), where $\hat{\sigma}$ comes from the full least-squares fit.

Why it does not prove our theorem

Part 1: yes, the construction needs n ≥ 2p, i.e. p ≤ n/2 counting all regressors. In our problem that means p + 1 ≤ n/2. Beyond that, the zero-row trick only works because the noise is Gaussian. The synthetic responses must have the same law as real noise, and an unknown exchangeable law cannot be sampled. The method also controls the false discovery rate over many variables rather than testing one coefficient with β free.

Part 2: it proves neither branch. Gaussian noise is a much stronger assumption than ours.

Paper 12 — Permuting the residual space (exact under spherical errors)

The papers

- Myung-Hoe Huh & Myoungshic Jhun, *Random permutation testing in multiple linear regression*, Communications in Statistics – Theory and Methods 30(10), 2001.
- Sara Kherad-Pajouh & Olivier Renaud, *An exact permutation method for testing any effect in balanced and unbalanced fixed effect ANOVA*, Computational Statistics & Data Analysis 54(7), 2010.

What they prove

Let Q ∈ ℝ^(n×(n−p)) have orthonormal columns orthogonal to the columns of X. Under H_0,

$$Q^\top Y = Q^\top X\beta + Q^\top \varepsilon = Q^\top \varepsilon \quad (\beta \text{ drops out exactly})$$

The test compares $T(Q^T Y, Q^T Z)$ with $T(P\, Q^T Y, Q^T Z)$ over permutations P of the $n - p$ coordinates. It is exact whenever **$Q^T\varepsilon$ is exchangeable**. That holds when ε is spherical, e.g. $N(0, \sigma^2 I)$ or any rotation-invariant law, because then $Q^T\varepsilon$ is spherical in $\mathbb{R}^{(n-p)}$.

Why it does not prove our theorem

Part 1: under exchangeability alone, $Q^T\varepsilon$ need not be exchangeable, so the method is not valid. Small example: $n = 3$, $X = \mathbf{1}$, $Q = [(1, -1, 0)/\sqrt{2}, (1, 1, -2)/\sqrt{6}]$. Let ε be a uniformly random permutation of $(1, 0, 0)$, which is exchangeable. $Q^T\varepsilon$ takes the three values

$$(0.707, 0.408), \qquad (-0.707, 0.408), \qquad (0, -0.816)$$

Swapping the two coordinates of the first value gives $(0.408, 0.707)$, which is not in the support. So $Q^T\varepsilon$ is not exchangeable.

This paper is still useful for the presentation. Under spherical errors it needs no $p < n/2$, only $n - p \geq 2$, and so does the classical t-test under Gaussian errors. **So the n/2 barrier is specific to the weak assumption "ε exchangeable".** It is not a barrier of the linear model itself.

Part 2: it proves neither branch. It works for $p \geq n/2$ only under a stronger assumption than ours.

Paper 13 — Freedman–Lane and its high-dimensional variants (no finite-sample guarantee)

The papers

- David Freedman & David Lane, *A nonstochastic interpretation of reported significance levels*, Journal of Business & Economic Statistics 1(4), 1983.
- Jesse Hemerik, Magne Thoresen & Livio Finos, *Permutation testing in high-dimensional linear models: an empirical investigation*, Journal of Statistical Computation and Simulation, 2021.

What they do

Freedman–Lane. Fit the null model, $\hat{\beta} = (X^T X)^{-1} X^T Y$ and $e = (I - H_X)Y$. Build permuted data and compare the t-statistic of Z on the real data with its values on the permuted data:

$$Y_\pi^* = X\hat{\beta} + P_\pi e$$

Under H_0, $e = (I - H_X)\varepsilon$. In general $P_\pi(I - H_X)\varepsilon$ does not have the same law as $(I - H_X)\varepsilon$, so the method is **not exact**. Its validity is asymptotic, with p fixed.

Hemerik–Thoresen–Finos. They compare Freedman–Lane-type schemes in high dimensions, with the nuisance estimated by methods such as the lasso. Type I error control is assessed **by simulation only**.

Why they do not prove our theorem

Part 1: they do not need $p < n/2$, but they also do not "work" in our sense. No finite-sample validity (V) is proved under exchangeable errors for any p. Freedman–Lane also needs $p < n$ for the least-squares fit.

Part 2: they prove neither branch. An empirical size close to α is not a proof of (V).

Paper 14 — Robust (studentised) permutation tests for regression coefficients

The paper

Cyrus DiCiccio & Joseph P. Romano, *Robust permutation tests for correlation and regression coefficients*, Journal of the American Statistical Association 112, 2017 (Stanford technical report 2015-15).

What it proves

Setting. (Y_i, X_i, Z_i) i.i.d. with $E(\varepsilon_i X_i) = 0$, and a studentised Wald statistic W_n for the tested coefficients.

Exact results.

- Permuting the tested regressor while keeping $(Y_i, \text{nuisance}_i)$ together is exact when the tested regressor is **independent of (Y, nuisance)**.
- Sign-flipping Y is exact when the regressors are independent of symmetric errors.

Asymptotic results (Theorems 3.2–3.5). Under the weaker null, the permutation distribution of W_n converges to χ^2_k, with p **fixed** as $n \to \infty$. This covers Freedman–Lane too.

Why it does not prove our theorem

Part 1: its exact versions use a different null and a random design. "Tested regressor independent of everything" is not "ε exchangeable with X fixed". Under our model it gives only asymptotic validity with p fixed, which says nothing about $p \geq n/2$.

Part 2: it proves neither branch.

Paper 15 — Sign-based exact inference with projection

The paper

Elise Coudin & Jean-Marie Dufour, *Finite-sample distribution-free inference in linear median regressions under heteroscedasticity and non-linear dependence of unknown form*, Econometrics Journal 12, 2009.

What it proves

Model. $y = X\beta + u$, where the errors form a **conditional mediangale**: each u_t has conditional median 0 given X and the past.

Proposition 3.1. Under that assumption, the signs $s(u_1), \ldots, s(u_n)$ are i.i.d. ± 1 with probability 1/2 each, conditional on X. So a joint statistic

$$DS(\beta_0, \Omega_n) = s(y - X\beta_0)^\top X\, \Omega_n\, X^\top s(y - X\beta_0)$$

is pivotal under H_0: $\beta = \beta_0$, and Monte Carlo critical values give an exact joint test.

Subvectors by projection. A confidence set for one coefficient is the projection of the exact joint confidence set. Coverage stays at least $1 - \alpha$, so the result is conservative.

Why it does not prove our theorem

Part 1: different error assumption, so not applicable. Exchangeable errors need not have median 0, and their signs need not be independent. Example: $\varepsilon_i = W + \xi_i$ with a common shock W is exchangeable, but all signs move together. The paper also states no condition relating p and n, and does not analyse when the projected test has power.

Part 2: it proves neither branch. Its projection idea is, however, the same device we use in the Part 2 section, applied there to an exact joint test that *is* valid under exchangeability (Paper 10).

Paper 16 — Invariance-based tests in high-dimensional regression

The paper

Wenxuan Guo & Panos Toulis, *Invariance-based inference in high-dimensional regression with finite-sample guarantees*, arXiv, December 2023.

What it proves

Global null H_0: $\beta = 0$ (all coefficients). Suppose ε =d gε given X for every g in a group, e.g. sign flips. Take the ridge statistic and random sign flips G_r:

$$t(y, X) = \|X\hat{\beta}_\lambda\|^2, \qquad \text{pval} = \frac{1}{R+1}\Big(1 + \sum_r \mathbf{1}\{t(G_r y, X) > t(y, X)\}\Big)$$

Theorem 1. Under the global null, $E_0\psi_\alpha \le \alpha$ for every α and n. There is no nuisance, because every coefficient is 0 under that null.

Partial null $\beta_S = 0$ (Theorem 7). Use t_A = ‖Ay‖² with A X_{S^c} = 0, and randomise the residuals. The size bound is

$$\mathbb{P}(\text{reject}) \le \alpha + O(1/R) + o(1)$$

It holds under independent, mean-zero, equal-variance, sign-symmetric errors with bounded fourth moments, p − |S| < n, and growth conditions.

Why it does not prove our theorem

Part 1: its exact result has no nuisance, and its nuisance result is only asymptotic. Our H_0 leaves β free, so it is a *partial* null. For partial nulls the paper gives only an asymptotic size guarantee, under independent symmetric errors rather than exchangeable ones.

Part 2: it proves neither branch.

Paper 17 — Exact tests with nuisance-annihilating contrasts (fixed effects)

The paper

Stanisław M. S. Halkiewicz, *Exact inference in fixed-effect regressions with concentrated identifying variation*, arXiv, August 2026.

What it proves

Model. $Y_i = x_i\beta + d_i^T\gamma + \varepsilon_i$, where d_i holds the fixed-effect dummies (the nuisance, of dimension d_n). The null is H_0: β = β_0 with γ free.

Contrasts (Definition 3.1). Choose nonzero q_1, …, q_C that are functions of the design only, with pairwise disjoint supports and

$$q_c^\top D = 0 \quad \Longrightarrow \quad q_c^\top (Y - x\beta_0) = q_c^\top \varepsilon \ \text{ under } H_0$$

Theorem 3.3. Assume the errors are *blockwise symmetric*: blocks are independent and each block is centrally symmetric. Then flipping the signs of $(q_c^T\varepsilon)_c$ is an exact symmetry, and the sign-flip p-value satisfies

$$\mathbb{P}(p \le \alpha \mid x, D) \le \alpha \quad \text{for every } n$$

There is no restriction on d_n relative to n. Proposition 3.7 replaces symmetry by pairwise exchangeability within antithetic pairs.

Why it does not prove our theorem

Part 1: no p < n/2 is needed, but the design must be special and the assumption is different.

- The main theorem assumes independence across blocks plus symmetry, not exchangeability.
- Under exchangeability alone, one version does go through. This is our observation, not stated in the paper. Take contrasts of the antithetic form $q = \Sigma_k (e_\{a_k\} - e_\{b_k\})$. Swapping $a_k \leftrightarrow b_k$ for every k is a permutation, so it flips the sign of $q^T\varepsilon$ without changing the joint law.

- Such a q must satisfy $q^T X = 0$, i.e. Σ_k X_{a_k} = Σ_k X_{b_k}. For a continuous design that is a measure-zero event. It holds for 0/1 dummy designs (cycles in a mobility graph).
- Like Paper 8, this is a method that escapes n/2 for **special discrete designs** only.

Part 2: it proves neither branch for general designs. It is consistent with our special-design counterexample.

Paper 18 — Mosaic permutation tests for panel data

The paper

Asher Spector, Rina Foygel Barber & Emmanuel Candès, *Mosaic inference on panel data*, arXiv, 2025.

What it proves

The method fits the nuisance separately on "tiles" of a panel and permutes residuals within tiles. Exact validity holds under a **local exchangeability** condition on the panel errors, i.e. exchangeability within each tile, for testing whether clusters are independent or for confidence intervals in panel regressions.

Why it does not prove our theorem

Part 1: different problem and different assumption. The error structure is a panel with tiles, not one exchangeable vector. The paper gives no result for one exchangeable vector with $p \geq n/2$.

Part 2: it proves neither branch.

Paper 19 — Limitations of randomization tests (an impossibility paper)

The paper

Deniz Dutz & Xinyi Zhang, *Limitations of randomization tests in finite samples*, arXiv, December 2025 (v2 April 2026).

What it proves

It gives a necessary and sufficient condition for a null hypothesis to admit a **randomization test**, i.e. a test that is exact because of a group invariance of the null. For one-sample problems:

- on continuous supports, common nulls such as "mean zero" admit no randomization test;
- with linear group actions, the admissible nulls are subsets of symmetric or Gaussian families.

Why it does not prove our theorem

Part 1: not applicable. It treats one-sample problems with no regression, no nuisance β and no p.

Part 2: it is the closest thing to a "necessity" result, but it does not give one.

- It rules out *randomization tests*, not all valid tests. Our necessity branch would need: "no test whatsoever satisfies (V) and (P) when $p \geq n/2$".
- Our null is "$b = 0$, β arbitrary, ε exchangeable", and it *does* admit randomization structure (permutations), so their impossibility results do not transfer.

Paper 20 — Survey: randomization inference

The paper

David M. Ritzwoller, Joseph P. Romano & Azeem M. Shaikh, *Randomization inference: theory and applications*, arXiv, 2024.

What it covers

A survey of when randomization and permutation tests are exact (the randomization hypothesis, i.e. group invariance of the null) and when they are only asymptotically valid. It covers studentised permutation tests and regression applications.

Why it does not prove our theorem

Part 1 and Part 2: it contains no result for p ≥ n/2 under exchangeable errors, and no necessity result. We used its reference list as a cross-check that no paper was missed.

Part 2: is p < n/2 necessary? No — two counterexamples

Short answer. No paper proves that p < n/2 is necessary, and it is **false**.

- **Counterexample A** is elementary and works for a special design, reaching p = n − 3.
- **Counterexample B** works for **almost every design**. It is a fully rigorous theorem at n = 40, α = 0.05 for every p ≤ 25, and at n = 20, α = 0.10 for every p ≤ 12. One step of that proof is a computer-checked exact rank certificate.
- Something *is* necessary, but much weaker: **p ≤ n − 2** (Proposition C).

Everything in this section is **our own analysis**. It is not in any paper we found, and it should be checked independently before the presentation.

Which reading of "works". No method can work for literally every design: if Z is a combination of the columns of X, then b is not identifiable at all. The meaningful readings are "for at least one design" (Counterexample A) and "for almost every design" (Counterexample B).

Counterexample A — repeated rows (any n ≥ 5, α = 1/24)

Design. Rows 1–4 share the same nuisance row: $X_1 = X_2 = X_3 = X_4$. The other rows are arbitrary. For example, X = [indicator of rows 1–4, e_5, …, e_n] has **p = n − 3 columns and rank n − 3**. Let $\tilde{z} = (Z_1, \ldots, Z_4)$ minus its mean. Assume its entries are distinct and no reordering of them equals $-\tilde{z}$ (true for almost every Z).

Test. G is the 24 permutations of rows 1–4 (all other rows fixed). Let $T(Y) = |\tilde{z}^{\mathsf{T}} Y_{1:4}|$ and

$$\hat{p} = \frac{\#\{g \in G : T(gY) \geq T(Y)\}}{24}, \qquad \text{reject } H_0 \text{ if } \hat{p} \leq \tfrac{1}{24} \ (\leq 0.05)$$

Validity. Under H_0, $Y_{1:4} = c\mathbf{1} + \varepsilon_{1:4}$ with $c = X_1^{\mathsf{T}}\beta$, and $\tilde{z}^{\mathsf{T}}\mathbf{1} = 0$, so $T(gY) = |\tilde{z}^{\mathsf{T}}(g\varepsilon)_{1:4}|$. Each g is a permutation of all n rows, so $g\varepsilon =_d \varepsilon$. The standard group argument then gives $P(\hat{p} \leq 1/24) \leq 1/24$, for every β and every exchangeable law.

Power. Under H_1,

$$T(gY) = \left| b\,\tilde{z}^{\top} g\tilde{z} + \tilde{z}^{\top}(g\varepsilon)_{1:4} \right|, \qquad T(Y) = \left| b\,\|\tilde{z}\|^2 + \tilde{z}^{\top}\varepsilon_{1:4} \right|$$

By Cauchy–Schwarz $|\tilde{z}^{\mathsf{T}} g\tilde{z}| < \|\tilde{z}\|^2$ for $g \neq \mathrm{id}$, so $\gamma := \|\tilde{z}\|^2 - \max_{g \neq \mathrm{id}} |\tilde{z}^{\mathsf{T}} g\tilde{z}| > 0$. Hence

$$T(Y) - T(gY) \ \geq \ |b|\,\gamma - 2\|\tilde{z}\|\,\|\varepsilon_{1:4}\| > 0 \quad \text{whenever} \quad |b| > 2\|\tilde{z}\|\,\|\varepsilon\|/\gamma$$

Then $\hat{p} = 1/24$ and the test rejects. So $P(\text{reject}) \geq P(\|\varepsilon\| < \gamma|b|/(2\|\tilde{z}\|)) \to 1$, uniformly in β. ∎

Matched-pairs version (the D'Haultfœuille–Tuvaandorj idea). n = 2m with X = pair indicators, so p = n/2. The test flips the signs of within-pair differences over all 2^m swaps. In our simulation, n = 20, p = 10, with errors a common Cauchy shock plus i.i.d. Cauchy (exchangeable, not independent, no moments): size 0.045, power 0.39 at b = 3 and 0.78 at b = 10 (4,000 replications).

Limitations. It needs repeated nuisance rows; for a continuous design it is the trivial test. With 4 shared rows the smallest level is 1/24. For a smaller α, use k shared rows with $k! \geq 1/\alpha$; the same proof applies, with rank n − k + 1. An independent simulation gave size 0.040 (bound 0.0417), and the power bound held in all 20,000 draws.

Counterexample B — almost every design

Idea. Take Kolumbán's exact *joint* test (Paper 10). Use the cyclic group instead of random permutations, and project it onto b = 0 (Paper 15's device). The n/2 barrier comes from comparing the data with one permuted copy at a time. Requiring $\mathbf{q \geq 2}$ copies to agree removes it.

Notation.

- $\Psi = [X, Z] \in \mathbb{R}^{n\times(p+1)}$ has full column rank, $p \leq n - 2$. $M = I - \Psi(\Psi^{T}\Psi)^{-1}\Psi^{T}$ and $V = \mathrm{col}(\Psi)$.
- $g_0 = \mathrm{id}, g_1, \ldots, g_{n-1}$ are the cyclic shifts. Take $\alpha = q/n$ with q a positive integer.

For each candidate $\beta' \in \mathbb{R}^p$ define

$$A(\beta') = \#\{\, k \in \{1, \ldots, n-1\} : \ \|Mg_k(Y - X\beta')\| \leq \|MY\| \,\}, \qquad \textbf{reject } H_0 \iff A(\beta') < q \ \textbf{ for every } \beta' \in \mathbb{R}^p$$

Theorem B1 (validity, every p and every design). $P_{\beta, b=0}(\text{reject}) \leq \alpha$ for every β and every exchangeable law of ε.

Proof. Rejecting needs $A(\beta) < q$ at the true β. Under H_0, $Y - X\beta = \varepsilon$ and $MY = M\varepsilon$. Put $t_k(\varepsilon) = \|Mg_k\varepsilon\|$, so $t_k(g_j\varepsilon) = t_{k+j}(\varepsilon)$ (indices mod n). Let $R(\varepsilon) = \#\{k : t_k \leq t_0\}$; rejection implies $R \leq q$. Since $g_j\varepsilon =_d \varepsilon$,

$$\mathbb{P}(R(\varepsilon) \leq q) = \frac{1}{n}\sum_{j=0}^{n-1} \mathbb{P}\big(\#\{k : t_k \leq t_j\} \leq q\big) = \frac{1}{n}\,\mathbb{E}\,\#\big\{j : \#\{k : t_k \leq t_j\} \leq q\big\} \leq \frac{q}{n} = \alpha$$

The last step holds because at most q of n numbers can have at most q numbers $\leq$ themselves. ∎

Theorem B2 (power). Suppose condition **(R_q)** holds: for every set S of q non-identity shifts, the stacked matrix $B_S = [Mg_k\Psi]_{k\in S} \in \mathbb{R}^{qn\times(p+1)}$ has full column rank. Let $\sigma = \min_S \sigma_{\min}(B_S) > 0$. Then, for every β and every realisation of ε,

$$|b| > \frac{2\sqrt{q}\,\|\varepsilon\|}{\sigma} \implies \textbf{the test rejects}, \qquad \text{so} \qquad \mathbb{P}(\text{reject}) \ \geq \ \mathbb{P}\Big(\|\varepsilon\| < \frac{\sigma|b|}{2\sqrt{q}}\Big) \ \to \ 1$$

The convergence is uniform in β.

Proof. Suppose $A(\beta') \geq q$, attained on a set S. Put $w = (\beta - \beta', b)$, so $Y - X\beta' = \Psi w + \varepsilon$. For $k \in S$,

$$\|Mg_k\Psi w\| \leq \|Mg_k(\Psi w + \varepsilon)\| + \|Mg_k\varepsilon\| \leq \|M\varepsilon\| + \|\varepsilon\| \leq 2\|\varepsilon\|$$

Therefore $\sigma|b| \leq \sigma\|w\| \leq \|B_S\, w\| \leq 2\sqrt{q}\|\varepsilon\|$. ∎

The bound is valid but loose. In an independent check (n = 20, p = 12, $\sigma = 0.037$), the test rejected 186 of 200 times at b = 100, where the guarantee itself only applies from about $b = 10^3$. The test is exactly the projection of Kolumbán et al.'s joint region (Paper 10) built with the cyclic group, so Theorem B1 is also a corollary of their exactness argument.

Lemma B3 (where n/2 comes from). B_S has full column rank if and only if

$$V \cap \bigcap_{k\in S} g_k^{-1}V = \{0\}, \qquad \textbf{and always} \qquad \dim\Big(V \cap \bigcap_{k\in S} g_k^{-1}V\Big) \ \geq \ (q+1)(p+1) - qn$$

So (R_q) needs $\mathbf{(q+1)(p+1) \leq qn}$.

- With $\mathbf{q = 1}$ this reads $p + 1 \leq n/2$. That is exactly the n/2 barrier, and it is Kolumbán's condition $Q_i \succ 0$.
- With $q = \alpha n$ it allows $p + 1 \leq \alpha n^2/(\alpha n + 1)$, i.e. p up to about $n - 1/\alpha - 1$.

Theorem B4 (almost every design; computer-certified). (R_2) holds for Lebesgue-almost every (X, Z) in two cases: n = 40 ($\alpha = 0.05$) with $p \leq 25$, and n = 20 ($\alpha = 0.10$) with $p \leq 12$. It also holds when X contains an intercept column; the kernel is then exactly the constants, which do not affect $A(\beta')$.

Proof.

1. For a fixed S, (R_q) says the integer block matrix $[[g_{k_1}\Psi, -\Psi, 0], [g_{k_2}\Psi, 0, -\Psi]]$ has full column rank 3(p+1). That holds iff some 3(p+1)-minor, a polynomial in the entries of Ψ, is nonzero.

2. We exhibited an integer Ψ for which every pair S (171 pairs at n = 20, 741 at n = 40) has full rank modulo the prime $2^{31} - 1$. That implies full rank over $\mathbb{Q}$.
3. A polynomial that is nonzero at one point vanishes only on a null set, and there are finitely many S.
4. Dropping columns preserves (R_q), so the result holds for all smaller p.
5. As a check, p = 26 at n = 40 fails for every design without an intercept (the block matrix is 80×81), as Lemma B3 predicts. With an intercept the barrier becomes $(q+1)p \le q(n-1)$, working modulo the constants, and an independent check certified p = 26 at n = 40 with an intercept. ∎

In the intercept case, σ in Theorem B2 is computed with the intercept column removed from Ψ; M is unchanged. The certificate was reproduced independently, with a different random integer witness and separately written code.

Consequence. At n = 40, $\alpha = 0.05$ and $p = 25 > n/2 = 20$, the test above is valid under every exchangeable law. For almost every design it has power $\to 1$ as $|b| \to \infty$. Hence **$p < n/2$ is not necessary**.

Simulations (exact evaluation; with q = 2, each candidate pair is a convex second-order cone problem).

Setting	Errors	Size (b = 0)	Power
n = 20, p = 10, 39 random permutations, α = 0.05	Gaussian	0/300	0.74 (b = 5), 1.00 (b = 20)
n = 20, p = 12, same	Gaussian	0/300	0.12 (b = 50), 0.80 (b = 200)
n = 20, p = 12, same	Cauchy shock + scale-mixed t_2	0/300	0.75 ($b = 10^3$), 1.00 ($b = 10^4$ and 10^5)
n = 20, p = 16 (Lemma B3 fails)	same	0/300	0 at every b up to 100
n = 40, p = 25, cyclic group	Gaussian	0/200	0.60 (b = 20), 1.00 (b = 100)
n = 40, p = 25, cyclic group	Cauchy shock + scale-mixed t_2	0/200	0.07 (b = 20), 0.50 (b = 100), 0.97 ($b = 10^3$)

Honest caveats.

1. Power tends to 1 but can need a large |b| near the boundary of Lemma B3, because σ is then small. At p = 10 the power is good; at p = 12 with n = 20 it is weak.
2. The test is very conservative: size 0 was observed in every setting.
3. The cost is C(n−1, q) convex problems per data set.
4. **What remains to prove** for a general theorem is the conjecture below. With larger groups (e.g. all of S_n, $q = \alpha \cdot n!$), Lemma B3 allows every $p \le n - 2$. That would match Proposition C exactly.

Conjecture: for every n and q, if $(q+1)(p+1) \le qn$, then (R_q) holds for the cyclic group and almost every (X, Z)

Proposition C — what is necessary: $p \le n - 2$

Claim. Let p = n − 1 with [X, Z] invertible. No test that is valid under i.i.d. $N(0, \sigma^2)$ errors for every σ (these laws are exchangeable) has power tending to 1 **uniformly in β**. Counterexamples A and B both have uniform power, so they meet this stronger standard.

Proof.

1. Let ν be the unit vector orthogonal to col(X). Write $Z = Xa + (\nu^T Z)\nu$ and $h(\beta, c) = P(\varphi(X\beta + c\nu + \xi) = 1)$ with $\xi \sim N(0, I)$. Uniform power means: for every $\delta > 0$ there is C with $h(\beta, c) \ge 1 - \delta$ for all β whenever $|c| > C$.
2. Under H_0 with $\sigma > 1$, write $\sigma\xi =_d \xi + \sqrt{(\sigma^2 - 1)}\eta$ with $\eta = X\gamma(\eta) + (\nu^T\eta)\nu$ independent. Then
$$\mathbb{P}_{\beta,\sigma}(\text{reject}) = \mathbb{E}_\eta\, h\big(\beta + \sqrt{\sigma^2 - 1}\,\gamma(\eta),\ \sqrt{\sigma^2 - 1}\,\nu^\top \eta\big) \ \ge\ (1 - \delta)\,\mathbb{P}\big(\sqrt{\sigma^2 - 1}\,|\nu^\top \eta| > C\big) \ \to\ 1 - \delta$$
3. As $\sigma \to \infty$ this exceeds α, which contradicts validity. ∎

This is the classical "zero residual degrees of freedom" obstruction. For power that is uniform in β, the true boundary therefore lies between n/2 (refuted) and n − 1 (impossible).

The uniformity matters. Under Gaussian errors with $p = n - 1$ there *is* a valid test whose power tends to 1 for each fixed β (pointed out by our independent check). Let $\zeta = \nu^T Z$ and reject when

$$|\nu^\top Y| > c\,\|Y - (\nu^\top Y/\zeta)Z\|$$

The right-hand side does not involve b, and Anderson's inequality bounds the size for every β and σ. Whether such pointwise power is possible under general exchangeable errors (e.g. with a common shock) is open. So Proposition C should always be quoted with "uniform in β".

Part 2 summary

Question	Status
Does any paper prove $p < n/2$ necessary?	No (Papers 1–20)
Is $p < n/2$ necessary for one design?	**No**: Counterexample A, $p = n - 3$
Is $p < n/2$ necessary for almost every design?	**No** at $n = 40$ ($p \le 25$) and $n = 20$ ($p \le 12$): Theorems B1–B4. All n: Conjecture
Is anything necessary?	$\mathbf{p \le n - 2}$ for power uniform in β: Proposition C. Not for pointwise power under Gaussian errors
Most useful next step	Prove the conjecture (genericity of (R_q)), and ask an RPT or RPS author whether this projection analysis is known

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Sources

Papers reviewed (numbering as above):

1. Wen, Wang & Wang, Residual permutation test for regression coefficient testing, Annals of Statistics 2025.
2. Guan, A conformal test of linear models via permutation-augmented regressions, Annals of Statistics 2024.
3. Li, Zhou & Zhang, Group permutation testing in linear model, arXiv 2026.
4. Fixed-row PALMRT (Wang, Wang, Wang & Zou, Statistics & Probability Letters 2025): see the link in Paper 4.
5. RobustPALMRT (Hilbert, MacEachern & Zhang, Statistics in Medicine 2025): see the link in Paper 5.
6. Lei & Bickel, An assumption-free exact test for fixed-design linear models with exchangeable errors, Biometrika 2021.
7. Pouliot, An exact t-test, working paper 2024.
8. D'Haultfœuille & Tuvaandorj, A robust permutation test for subvector inference in linear regressions, Quantitative Economics 2024.
9. Guo, Toulis & Wang, Permutation inference under multi-way clustering and missing data, arXiv 2026.
10. Kolumbán, Vajk & Schoukens, Perturbed datasets methods for hypothesis testing, arXiv 2014. Szentpéteri & Csáji, Finite-sample identification of linear regression models with residual-permuted sums, arXiv 2024.
11. Barber & Candès, Controlling the false discovery rate via knockoffs, Annals of Statistics 2015.
12. Huh & Jhun, Communications in Statistics 2001. Kherad-Pajouh & Renaud, Computational Statistics & Data Analysis 2010.
13. Freedman & Lane, JBES 1983. Hemerik, Thoresen & Finos, Permutation testing in high-dimensional linear models, 2021.
14. DiCiccio & Romano, JASA 2017 (technical report).
15. Coudin & Dufour, Econometrics Journal 2009.
16. Guo & Toulis, Invariance-based inference in high-dimensional regression, arXiv 2023.
17. Halkiewicz, Exact inference in fixed-effect regressions with concentrated identifying variation, arXiv 2026.
18. Spector, Barber & Candès, Mosaic inference on panel data, arXiv 2025.
19. Dutz & Zhang, Limitations of randomization tests in finite samples, arXiv 2025.
20. Ritzwoller, Romano & Shaikh, Randomization inference: theory and applications, arXiv 2024.

Our own computations (Python scripts, available on request): rank checks for CPT and RPT; the matched-pairs simulation; the exact projection-test evaluations (one second-order cone problem per pair of permutations); and the modular-arithmetic rank certificate behind Theorem B4.

Appendix: Search log

The search ran on 19 and 26 October 2026 and covered arXiv postings up to 20 October 2026. Every candidate paper was opened in full text before being classified.

Citation chains (forward)

These list every indexed paper citing each key result. Each one was read at least at abstract level, and every relevant one was read in full.

Seed paper	Index used	Citing papers found	Any that answer our question?
CPT (Lei & Bickel 2021)	Semantic Scholar + OpenAlex	24 + 12 (overlapping)	No
RPT (Wen, Wang & Wang 2025)	Semantic Scholar + OpenAlex + Google Scholar	6 + 4 + 4 shown	No
PALMRT (Guan 2024)	Semantic Scholar + OpenAlex	11 + 1	No
D'Haultfœuille & Tuvaandorj 2024	Semantic Scholar + OpenAlex	12 + 9	No
Group PALMRT (Jan 2026)	Semantic Scholar	1	No
Fixed-row PALMRT, RobustPALMRT	Semantic Scholar + OpenAlex	0	—

Keyword searches

- **arXiv full search, sorted newest first:** about 25 queries such as "exchangeable errors", "residual permutation", "cyclic permutation test", "permutation linear model exact test", "randomization test regression" and "exact test exchangeable".
- **OpenAlex title and abstract search from 2015:** 17 phrase queries, including "exchangeable errors", "exchangeable noise", "permutation-augmented", "assumption-free linear model test" and "exact inference exchangeable regression".
- **Web search, standard and extended:** more than 40 queries, including impossibility and necessity phrasings ("n ≥ 2p", "p ≥ n/2", "lower bound", "any valid test"), theses, e-values, and conference venues.
- **Reference lists:** the related-work sections of the four 2026 papers (Group PALMRT, multi-way clustering, the fixed-effects paper and Dutz & Zhang), the Ritzwoller, Romano & Shaikh 2024 survey, and CPT's century-long review in its Appendix B.

What could not be fully checked

- Google Scholar flagged automated traffic and then returned degraded, date-filtered results. Its citation lists were only partly usable and matched the Semantic Scholar lists where visible.
- Semantic Scholar's keyword-search endpoint was rate-limited; its citation endpoint worked.
- The full text of Jouneau-Sion & Torrès (2014) could not be opened.
- For Pouliot, the 2024 working paper was read rather than the 2026 published version.
- Non-English literature, theses that are not indexed, and work under review that has not been posted cannot be searched.
- Two 2026 papers on random rotation-invariant designs (Xia et al.; Zhang, Ding, Zhou & Shi), noted in an earlier pass, could not be re-located on 26 October. They were dropped from the review rather than cited unverified. Their setting is a random design, so they would not answer our fixed-design question either way.
- For the projection test in Part 2, targeted searches for "projection of sign-perturbed sums / perturbed-dataset regions" and nuisance parameters found no analysis of power when p ≥ n/2.

Residual risk is low but not zero. The cheapest way to close it is a short email to an RPT author asking whether anything on $n/2 \le p < n$ under exchangeability exists or is in progress.